

PANCAKES AND TERNARY STRINGS

MATH 300

Problem 1: Lazy Caterer Problem: Let c_n denote the maximum number of pieces a circular pancake can be cut into with n cuts. What are c_0, c_1, c_2, c_3, c_4 ? By looking at the differences $c_n - c_{n-1}$, make a hypothesis for a recurrence relation for c_n .

Problem 2: How many ternary strings of length n that contain 00 are? Call this x_n .

- (1) Consider three distinct ways a length n binary string could start: 1, 2, 01, 02, 00. For each of these cases, how many ways are there to fill in the rest of the $n - 1$ or $n - 2$ positions in the string?
- (2) One of these five cases is not like the others. Which one?
- (3) Use the above answers to write x_n as a second order linear non-homogenous recurrence relation? What are the initial conditions?